1 代码运行结果

表 1: 按 cell_type 划分目标域数据集 [0.14:0.14:0.72]

脚本	训练集 R2	验证集
train_balanced_sampling.py（端到端训练 HGAT）	0.965	0.855
train_balanced_sampling.py（冻结 HGAT）	0.964	0.858
train_balanced_sampling_sep_mlp.py（端到端训练 HGAT）	0.949	0.853
train_balanced_sampling_sep_mlp.py（冻结 HGAT）	0.947	0.831
train_balanced_sampling_sep_mlp_disentangle_domain.py（端到端训练 HGAT）	0.972	0.824
train_balanced_sampling_sep_mlp_disentangle_domain.py（冻结 HGAT）	0.973	0.821

表 2: 随机划分目标域数据集 [0.14:0.14:0.72]

脚本	训练集 R2	验证集
train_balanced_sampling.py（冻结 HGAT）	0.964	0.858
train_balanced_sampling_sep_mlp.py（冻结 HGAT）	0.947	0.831
train_balanced_sampling_sep_mlp_disentangle_domain.py（冻结 HGAT）	0.973	0.821

1.1 train_balanced_sampling.py（端到端训练 HGAT）

```
e0, train_loss:0.8055, r2:-0.042, val_loss:1.0283, val_r2:-0.013
Model successfully saved
e1, train_loss:0.7879, r2:-0.006, val_loss:0.9965, val_r2:0.018
Model successfully saved
e2, train_loss:0.7616, r2:0.026, val_loss:0.9670, val_r2:0.047
Model successfully saved
e3, train_loss:0.7286, r2:0.055, val_loss:0.9389, val_r2:0.075
Model successfully saved
```

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e4, train_loss:0.6959, r2:0.082, val_loss:0.9117, val_r2:0.102
Model successfully saved
e5, train_loss:0.6758, r2:0.108, val_loss:0.8849, val_r2:0.128
Model successfully saved
e6, train_loss:0.6607, r2:0.134, val_loss:0.8579, val_r2:0.155
Model successfully saved
e7, train_loss:0.6406, r2:0.161, val_loss:0.8302, val_r2:0.182
Model successfully saved
e8, train_loss:0.6302, r2:0.188, val_loss:0.8019, val_r2:0.210
Model successfully saved
e9, train_loss:0.5898, r2:0.215, val_loss:0.7731, val_r2:0.238
Model successfully saved
e10, train_loss:0.5726, r2:0.244, val_loss:0.7438, val_r2:0.267
Model successfully saved
e11, train_loss:0.5644, r2:0.272, val_loss:0.7148, val_r2:0.296
Model successfully saved
e12, train_loss:0.5254, r2:0.302, val_loss:0.6860, val_r2:0.324
Model successfully saved
e13, train_loss:0.5060, r2:0.332, val_loss:0.6576, val_r2:0.352
Model successfully saved
e14, train_loss:0.4775, r2:0.362, val_loss:0.6284, val_r2:0.381
Model successfully saved
e15, train_loss:0.4579, r2:0.393, val_loss:0.5992, val_r2:0.410
Model successfully saved
e16, train_loss:0.4354, r2:0.425, val_loss:0.5695, val_r2:0.439
Model successfully saved
e17, train_loss:0.4208, r2:0.457, val_loss:0.5395, val_r2:0.469
Model successfully saved
e18, train_loss:0.3975, r2:0.489, val_loss:0.5095, val_r2:0.498
Model successfully saved
e19, train_loss:0.3947, r2:0.521, val_loss:0.4792, val_r2:0.528
Model successfully saved
e20, train_loss:0.3610, r2:0.553, val_loss:0.4478, val_r2:0.559
Model successfully saved
e21, train_loss:0.3458, r2:0.584, val_loss:0.4166, val_r2:0.590
Model successfully saved
e22, train_loss:0.3455, r2:0.613, val_loss:0.3877, val_r2:0.618
Model successfully saved
e23, train_loss:0.3561, r2:0.641, val_loss:0.3640, val_r2:0.641
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Model successfully saved
e24, train_loss:0.3269, r2:0.665, val_loss:0.3445, val_r2:0.661
Model successfully saved
e25, train_loss:0.3260, r2:0.687, val_loss:0.3289, val_r2:0.676
Model successfully saved
e26, train_loss:0.3110, r2:0.706, val_loss:0.3175, val_r2:0.687
Model successfully saved
e27, train_loss:0.2995, r2:0.722, val_loss:0.3080, val_r2:0.697
Model successfully saved
e28, train_loss:0.2945, r2:0.737, val_loss:0.2986, val_r2:0.706
Model successfully saved
e29, train_loss:0.2724, r2:0.750, val_loss:0.2876, val_r2:0.717
Model successfully saved
e30, train_loss:0.2957, r2:0.763, val_loss:0.2759, val_r2:0.728
Model successfully saved
e31, train_loss:0.2727, r2:0.775, val_loss:0.2634, val_r2:0.740
Model successfully saved
e32, train_loss:0.2609, r2:0.787, val_loss:0.2524, val_r2:0.751
Model successfully saved
e33, train_loss:0.2187, r2:0.798, val_loss:0.2425, val_r2:0.761
Model successfully saved
e34, train_loss:0.2491, r2:0.808, val_loss:0.2340, val_r2:0.770
Model successfully saved
e35, train_loss:0.2281, r2:0.818, val_loss:0.2266, val_r2:0.777
Model successfully saved
e36, train_loss:0.2269, r2:0.828, val_loss:0.2210, val_r2:0.782
Model successfully saved
e37, train_loss:0.2000, r2:0.837, val_loss:0.2156, val_r2:0.788
Model successfully saved
e38, train_loss:0.1963, r2:0.845, val_loss:0.2093, val_r2:0.794
Model successfully saved
e39, train_loss:0.1992, r2:0.853, val_loss:0.2028, val_r2:0.800
Model successfully saved
e40, train_loss:0.1900, r2:0.861, val_loss:0.1952, val_r2:0.808
Model successfully saved
e41, train_loss:0.1903, r2:0.869, val_loss:0.1883, val_r2:0.814
Model successfully saved
e42, train_loss:0.1940, r2:0.876, val_loss:0.1824, val_r2:0.820
Model successfully saved
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e43, train_loss:0.1980, r2:0.883, val_loss:0.1777, val_r2:0.825
Model successfully saved
e44, train_loss:0.1822, r2:0.888, val_loss:0.1730, val_r2:0.830
Model successfully saved
e45, train_loss:0.1678, r2:0.894, val_loss:0.1691, val_r2:0.833
Model successfully saved
e46, train_loss:0.1714, r2:0.898, val_loss:0.1667, val_r2:0.836
Model successfully saved
e47, train_loss:0.1880, r2:0.903, val_loss:0.1663, val_r2:0.836
Model successfully saved
e48, train_loss:0.1508, r2:0.906, val_loss:0.1648, val_r2:0.838
Model successfully saved
e49, train_loss:0.1579, r2:0.910, val_loss:0.1632, val_r2:0.839
Model successfully saved
e50, train_loss:0.1632, r2:0.914, val_loss:0.1621, val_r2:0.840
Model successfully saved
e51, train_loss:0.1526, r2:0.918, val_loss:0.1607, val_r2:0.842
Model successfully saved
e52, train_loss:0.1593, r2:0.921, val_loss:0.1591, val_r2:0.843
Model successfully saved
e53, train_loss:0.1597, r2:0.925, val_loss:0.1582, val_r2:0.844
Model successfully saved
e54, train_loss:0.1602, r2:0.929, val_loss:0.1583, val_r2:0.844
e55, train_loss:0.1531, r2:0.932, val_loss:0.1592, val_r2:0.843
e56, train_loss:0.1493, r2:0.934, val_loss:0.1598, val_r2:0.843
e57, train_loss:0.1531, r2:0.936, val_loss:0.1593, val_r2:0.843
e58, train_loss:0.1577, r2:0.939, val_loss:0.1587, val_r2:0.844
e59, train_loss:0.1337, r2:0.942, val_loss:0.1571, val_r2:0.845
Model successfully saved
e60, train_loss:0.1543, r2:0.944, val_loss:0.1550, val_r2:0.847
Model successfully saved
e61, train_loss:0.1493, r2:0.946, val_loss:0.1542, val_r2:0.848
Model successfully saved
e62, train_loss:0.1500, r2:0.948, val_loss:0.1536, val_r2:0.849
Model successfully saved
e63, train_loss:0.1368, r2:0.950, val_loss:0.1540, val_r2:0.848
e64, train_loss:0.1366, r2:0.952, val_loss:0.1552, val_r2:0.847
e65, train_loss:0.1258, r2:0.953, val_loss:0.1549, val_r2:0.847
e66, train_loss:0.1306, r2:0.954, val_loss:0.1538, val_r2:0.848
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e67, train_loss:0.1515, r2:0.956, val_loss:0.1547, val_r2:0.848
e68, train_loss:0.1400, r2:0.957, val_loss:0.1554, val_r2:0.847
e69, train_loss:0.1292, r2:0.958, val_loss:0.1552, val_r2:0.847
e70, train_loss:0.1437, r2:0.959, val_loss:0.1559, val_r2:0.846
e71, train_loss:0.1211, r2:0.960, val_loss:0.1549, val_r2:0.847
e72, train_loss:0.1182, r2:0.961, val_loss:0.1520, val_r2:0.850
Model successfully saved
e73, train_loss:0.1327, r2:0.962, val_loss:0.1496, val_r2:0.853
Model successfully saved
e74, train_loss:0.1287, r2:0.964, val_loss:0.1482, val_r2:0.854
Model successfully saved
e75, train_loss:0.1205, r2:0.965, val_loss:0.1475, val_r2:0.855
Model successfully saved
e76, train_loss:0.1174, r2:0.966, val_loss:0.1479, val_r2:0.854
e77, train_loss:0.1162, r2:0.967, val_loss:0.1490, val_r2:0.853
e78, train_loss:0.1303, r2:0.967, val_loss:0.1513, val_r2:0.851
e79, train_loss:0.1271, r2:0.967, val_loss:0.1542, val_r2:0.848
e80, train_loss:0.1239, r2:0.968, val_loss:0.1570, val_r2:0.845
e81, train_loss:0.1247, r2:0.968, val_loss:0.1570, val_r2:0.845
e82, train_loss:0.1224, r2:0.969, val_loss:0.1565, val_r2:0.846
e83, train_loss:0.1298, r2:0.970, val_loss:0.1555, val_r2:0.847
e84, train_loss:0.1181, r2:0.971, val_loss:0.1550, val_r2:0.847
e85, train_loss:0.1136, r2:0.972, val_loss:0.1553, val_r2:0.847
e86, train_loss:0.1205, r2:0.972, val_loss:0.1564, val_r2:0.846
e87, train_loss:0.1130, r2:0.973, val_loss:0.1568, val_r2:0.845
e88, train_loss:0.1356, r2:0.973, val_loss:0.1594, val_r2:0.843
e89, train_loss:0.1227, r2:0.974, val_loss:0.1612, val_r2:0.841
e90, train_loss:0.1143, r2:0.974, val_loss:0.1615, val_r2:0.841
e91, train_loss:0.1135, r2:0.975, val_loss:0.1620, val_r2:0.840
e92, train_loss:0.1192, r2:0.976, val_loss:0.1633, val_r2:0.839
e93, train_loss:0.1146, r2:0.976, val_loss:0.1648, val_r2:0.838
e94, train_loss:0.1183, r2:0.977, val_loss:0.1666, val_r2:0.836
e95, train_loss:0.1125, r2:0.977, val_loss:0.1675, val_r2:0.835
e96, train_loss:0.1108, r2:0.977, val_loss:0.1684, val_r2:0.834
e97, train_loss:0.1110, r2:0.978, val_loss:0.1682, val_r2:0.834
e98, train_loss:0.1126, r2:0.978, val_loss:0.1678, val_r2:0.835
e99, train_loss:0.1140, r2:0.979, val_loss:0.1674, val_r2:0.835
```

1.2 train_balanced_sampling_sep_mlp.py（端到端训练 HGAT）

```
e0, train_loss:0.8065, r2:-0.042, val_loss:1.0302, val_r2:-0.015
Model successfully saved
e1, train_loss:0.7575, r2:-0.007, val_loss:1.0000, val_r2:0.015
Model successfully saved
e2, train_loss:0.7421, r2:0.025, val_loss:0.9718, val_r2:0.043
Model successfully saved
e3, train_loss:0.7047, r2:0.054, val_loss:0.9447, val_r2:0.069
Model successfully saved
e4, train_loss:0.6746, r2:0.082, val_loss:0.9182, val_r2:0.095
Model successfully saved
e5, train_loss:0.6534, r2:0.108, val_loss:0.8920, val_r2:0.121
Model successfully saved
e6, train_loss:0.6361, r2:0.135, val_loss:0.8652, val_r2:0.148
Model successfully saved
e7, train_loss:0.6357, r2:0.161, val_loss:0.8378, val_r2:0.175
Model successfully saved
e8, train_loss:0.6008, r2:0.188, val_loss:0.8093, val_r2:0.203
Model successfully saved
e9, train_loss:0.5859, r2:0.216, val_loss:0.7800, val_r2:0.232
Model successfully saved
e10, train_loss:0.5564, r2:0.245, val_loss:0.7495, val_r2:0.262
Model successfully saved
e11, train_loss:0.5337, r2:0.275, val_loss:0.7182, val_r2:0.292
Model successfully saved
e12, train_loss:0.5228, r2:0.306, val_loss:0.6860, val_r2:0.324
Model successfully saved
e13, train_loss:0.4824, r2:0.338, val_loss:0.6529, val_r2:0.357
Model successfully saved
e14, train_loss:0.4737, r2:0.371, val_loss:0.6186, val_r2:0.391
Model successfully saved
e15, train_loss:0.4489, r2:0.405, val_loss:0.5837, val_r2:0.425
Model successfully saved
e16, train_loss:0.4393, r2:0.440, val_loss:0.5482, val_r2:0.460
Model successfully saved
e17, train_loss:0.4133, r2:0.476, val_loss:0.5124, val_r2:0.495
Model successfully saved
e18, train_loss:0.3902, r2:0.512, val_loss:0.4766, val_r2:0.531
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Model successfully saved
e19, train_loss:0.3712, r2:0.547, val_loss:0.4412, val_r2:0.565
Model successfully saved
e20, train_loss:0.3387, r2:0.583, val_loss:0.4066, val_r2:0.599
Model successfully saved
e21, train_loss:0.3377, r2:0.617, val_loss:0.3733, val_r2:0.632
Model successfully saved
e22, train_loss:0.3264, r2:0.650, val_loss:0.3417, val_r2:0.663
Model successfully saved
e23, train_loss:0.3002, r2:0.682, val_loss:0.3125, val_r2:0.692
Model successfully saved
e24, train_loss:0.2832, r2:0.710, val_loss:0.2863, val_r2:0.718
Model successfully saved
e25, train_loss:0.2754, r2:0.737, val_loss:0.2635, val_r2:0.740
Model successfully saved
e26, train_loss:0.2535, r2:0.759, val_loss:0.2446, val_r2:0.759
Model successfully saved
e27, train_loss:0.2359, r2:0.779, val_loss:0.2299, val_r2:0.774
Model successfully saved
e28, train_loss:0.2312, r2:0.794, val_loss:0.2194, val_r2:0.784
Model successfully saved
e29, train_loss:0.2237, r2:0.806, val_loss:0.2127, val_r2:0.790
Model successfully saved
e30, train_loss:0.2251, r2:0.815, val_loss:0.2091, val_r2:0.794
Model successfully saved
e31, train_loss:0.2164, r2:0.822, val_loss:0.2077, val_r2:0.795
Model successfully saved
e32, train_loss:0.2099, r2:0.827, val_loss:0.2076, val_r2:0.796
Model successfully saved
e33, train_loss:0.2080, r2:0.832, val_loss:0.2077, val_r2:0.795
e34, train_loss:0.2008, r2:0.838, val_loss:0.2076, val_r2:0.795
e35, train_loss:0.1994, r2:0.845, val_loss:0.2070, val_r2:0.796
Model successfully saved
e36, train_loss:0.2011, r2:0.852, val_loss:0.2057, val_r2:0.797
Model successfully saved
e37, train_loss:0.1850, r2:0.861, val_loss:0.2039, val_r2:0.799
Model successfully saved
e38, train_loss:0.1878, r2:0.869, val_loss:0.2019, val_r2:0.801
Model successfully saved
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e39, train_loss:0.1819, r2:0.878, val_loss:0.1996, val_r2:0.803
Model successfully saved
e40, train_loss:0.1768, r2:0.886, val_loss:0.1969, val_r2:0.806
Model successfully saved
e41, train_loss:0.1646, r2:0.894, val_loss:0.1939, val_r2:0.809
Model successfully saved
e42, train_loss:0.1557, r2:0.900, val_loss:0.1905, val_r2:0.812
Model successfully saved
e43, train_loss:0.1624, r2:0.906, val_loss:0.1865, val_r2:0.816
Model successfully saved
e44, train_loss:0.1510, r2:0.911, val_loss:0.1821, val_r2:0.821
Model successfully saved
e45, train_loss:0.1569, r2:0.916, val_loss:0.1774, val_r2:0.825
Model successfully saved
e46, train_loss:0.1543, r2:0.921, val_loss:0.1726, val_r2:0.830
Model successfully saved
e47, train_loss:0.1448, r2:0.925, val_loss:0.1678, val_r2:0.835
Model successfully saved
e48, train_loss:0.1452, r2:0.929, val_loss:0.1635, val_r2:0.839
Model successfully saved
e49, train_loss:0.1440, r2:0.933, val_loss:0.1597, val_r2:0.843
Model successfully saved
e50, train_loss:0.1464, r2:0.937, val_loss:0.1563, val_r2:0.846
Model successfully saved
e51, train_loss:0.1320, r2:0.940, val_loss:0.1536, val_r2:0.849
Model successfully saved
e52, train_loss:0.1316, r2:0.944, val_loss:0.1515, val_r2:0.851
Model successfully saved
e53, train_loss:0.1330, r2:0.946, val_loss:0.1502, val_r2:0.852
Model successfully saved
e54, train_loss:0.1351, r2:0.949, val_loss:0.1495, val_r2:0.853
Model successfully saved
e55, train_loss:0.1229, r2:0.951, val_loss:0.1496, val_r2:0.853
e56, train_loss:0.1388, r2:0.953, val_loss:0.1504, val_r2:0.852
e57, train_loss:0.1320, r2:0.955, val_loss:0.1518, val_r2:0.851
e58, train_loss:0.1310, r2:0.957, val_loss:0.1537, val_r2:0.849
e59, train_loss:0.1178, r2:0.959, val_loss:0.1560, val_r2:0.846
e60, train_loss:0.1224, r2:0.961, val_loss:0.1587, val_r2:0.844
e61, train_loss:0.1234, r2:0.963, val_loss:0.1617, val_r2:0.841
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e62, train_loss:0.1319, r2:0.964, val_loss:0.1648, val_r2:0.838
e63, train_loss:0.1215, r2:0.966, val_loss:0.1680, val_r2:0.835
e64, train_loss:0.1185, r2:0.967, val_loss:0.1710, val_r2:0.832
e65, train_loss:0.1198, r2:0.968, val_loss:0.1737, val_r2:0.829
e66, train_loss:0.1156, r2:0.969, val_loss:0.1761, val_r2:0.827
e67, train_loss:0.1219, r2:0.970, val_loss:0.1779, val_r2:0.825
e68, train_loss:0.1120, r2:0.971, val_loss:0.1793, val_r2:0.823
e69, train_loss:0.1097, r2:0.972, val_loss:0.1800, val_r2:0.823
e70, train_loss:0.1071, r2:0.972, val_loss:0.1803, val_r2:0.822
e71, train_loss:0.1240, r2:0.973, val_loss:0.1801, val_r2:0.823
e72, train_loss:0.1163, r2:0.974, val_loss:0.1797, val_r2:0.823
e73, train_loss:0.1214, r2:0.975, val_loss:0.1791, val_r2:0.824
e74, train_loss:0.1101, r2:0.975, val_loss:0.1785, val_r2:0.824
e75, train_loss:0.1062, r2:0.976, val_loss:0.1778, val_r2:0.825
e76, train_loss:0.1094, r2:0.976, val_loss:0.1773, val_r2:0.825
e77, train_loss:0.1087, r2:0.977, val_loss:0.1770, val_r2:0.826
e78, train_loss:0.1066, r2:0.977, val_loss:0.1770, val_r2:0.826
e79, train_loss:0.1050, r2:0.977, val_loss:0.1772, val_r2:0.825
e80, train_loss:0.1066, r2:0.978, val_loss:0.1777, val_r2:0.825
e81, train_loss:0.1098, r2:0.978, val_loss:0.1784, val_r2:0.824
e82, train_loss:0.0998, r2:0.979, val_loss:0.1794, val_r2:0.823
e83, train_loss:0.1085, r2:0.979, val_loss:0.1805, val_r2:0.822
e84, train_loss:0.1092, r2:0.979, val_loss:0.1817, val_r2:0.821
e85, train_loss:0.1060, r2:0.980, val_loss:0.1830, val_r2:0.820
e86, train_loss:0.1055, r2:0.980, val_loss:0.1844, val_r2:0.818
e87, train_loss:0.1105, r2:0.980, val_loss:0.1857, val_r2:0.817
e88, train_loss:0.1104, r2:0.981, val_loss:0.1868, val_r2:0.816
e89, train_loss:0.1035, r2:0.981, val_loss:0.1876, val_r2:0.815
e90, train_loss:0.0951, r2:0.981, val_loss:0.1882, val_r2:0.815
e91, train_loss:0.0999, r2:0.981, val_loss:0.1885, val_r2:0.814
e92, train_loss:0.1029, r2:0.982, val_loss:0.1886, val_r2:0.814
e93, train_loss:0.1006, r2:0.982, val_loss:0.1886, val_r2:0.814
e94, train_loss:0.1111, r2:0.982, val_loss:0.1885, val_r2:0.814
e95, train_loss:0.1150, r2:0.983, val_loss:0.1883, val_r2:0.814
e96, train_loss:0.1031, r2:0.983, val_loss:0.1882, val_r2:0.815
e97, train_loss:0.1054, r2:0.983, val_loss:0.1880, val_r2:0.815
e98, train_loss:0.1131, r2:0.983, val_loss:0.1878, val_r2:0.815
e99, train_loss:0.1093, r2:0.984, val_loss:0.1876, val_r2:0.815
```

1.3 train_balanced_sampling_sep_mlp_disentangle_domain.py（端到端训练 HGAT）

```
e0, train_loss:0.7745, r2:-0.006, val_loss:1.0036, val_r2:0.011
Model successfully saved
e1, train_loss:0.7374, r2:0.029, val_loss:0.9751, val_r2:0.039
Model successfully saved
e2, train_loss:0.7284, r2:0.061, val_loss:0.9476, val_r2:0.067
Model successfully saved
e3, train_loss:0.6925, r2:0.092, val_loss:0.9197, val_r2:0.094
Model successfully saved
e4, train_loss:0.6524, r2:0.124, val_loss:0.8894, val_r2:0.124
Model successfully saved
e5, train_loss:0.6485, r2:0.157, val_loss:0.8558, val_r2:0.157
Model successfully saved
e6, train_loss:0.6082, r2:0.194, val_loss:0.8195, val_r2:0.193
Model successfully saved
e7, train_loss:0.5922, r2:0.233, val_loss:0.7805, val_r2:0.231
Model successfully saved
e8, train_loss:0.5461, r2:0.274, val_loss:0.7387, val_r2:0.272
Model successfully saved
e9, train_loss:0.5237, r2:0.319, val_loss:0.6937, val_r2:0.317
Model successfully saved
e10, train_loss:0.5139, r2:0.365, val_loss:0.6463, val_r2:0.363
Model successfully saved
e11, train_loss:0.4559, r2:0.413, val_loss:0.5969, val_r2:0.412
Model successfully saved
e12, train_loss:0.4395, r2:0.462, val_loss:0.5468, val_r2:0.461
Model successfully saved
e13, train_loss:0.4196, r2:0.510, val_loss:0.4966, val_r2:0.511
Model successfully saved
e14, train_loss:0.3742, r2:0.557, val_loss:0.4482, val_r2:0.558
Model successfully saved
e15, train_loss:0.3628, r2:0.601, val_loss:0.4035, val_r2:0.603
Model successfully saved
e16, train_loss:0.3431, r2:0.641, val_loss:0.3645, val_r2:0.641
Model successfully saved
e17, train_loss:0.3211, r2:0.678, val_loss:0.3316, val_r2:0.673
Model successfully saved
```

```
e18, train_loss:0.2987, r2:0.711, val_loss:0.3050, val_r2:0.700
Model successfully saved
e19, train_loss:0.2923, r2:0.741, val_loss:0.2845, val_r2:0.720
Model successfully saved
e20, train_loss:0.2666, r2:0.767, val_loss:0.2687, val_r2:0.735
Model successfully saved
e21, train_loss:0.2407, r2:0.789, val_loss:0.2558, val_r2:0.748
Model successfully saved
e22, train_loss:0.2505, r2:0.807, val_loss:0.2439, val_r2:0.760
Model successfully saved
e23, train_loss:0.2406, r2:0.823, val_loss:0.2314, val_r2:0.772
Model successfully saved
e24, train_loss:0.2076, r2:0.838, val_loss:0.2158, val_r2:0.787
Model successfully saved
e25, train_loss:0.2099, r2:0.854, val_loss:0.2019, val_r2:0.801
Model successfully saved
e26, train_loss:0.1825, r2:0.869, val_loss:0.1969, val_r2:0.806
Model successfully saved
e27, train_loss:0.1965, r2:0.881, val_loss:0.2010, val_r2:0.802
e28, train_loss:0.1794, r2:0.890, val_loss:0.2081, val_r2:0.795
e29, train_loss:0.1739, r2:0.899, val_loss:0.2124, val_r2:0.791
e30, train_loss:0.1701, r2:0.908, val_loss:0.2157, val_r2:0.788
e31, train_loss:0.1665, r2:0.915, val_loss:0.2205, val_r2:0.783
e32, train_loss:0.1596, r2:0.917, val_loss:0.2263, val_r2:0.777
e33, train_loss:0.1623, r2:0.921, val_loss:0.2335, val_r2:0.770
e34, train_loss:0.1705, r2:0.926, val_loss:0.2429, val_r2:0.761
e35, train_loss:0.1492, r2:0.931, val_loss:0.2506, val_r2:0.753
e36, train_loss:0.1660, r2:0.934, val_loss:0.2498, val_r2:0.754
e37, train_loss:0.1509, r2:0.937, val_loss:0.2409, val_r2:0.763
e38, train_loss:0.1502, r2:0.942, val_loss:0.2306, val_r2:0.773
e39, train_loss:0.1447, r2:0.945, val_loss:0.2234, val_r2:0.780
e40, train_loss:0.1350, r2:0.947, val_loss:0.2185, val_r2:0.785
e41, train_loss:0.1460, r2:0.949, val_loss:0.2152, val_r2:0.788
e42, train_loss:0.1359, r2:0.953, val_loss:0.2124, val_r2:0.791
e43, train_loss:0.1339, r2:0.955, val_loss:0.2079, val_r2:0.795
e44, train_loss:0.1369, r2:0.957, val_loss:0.2013, val_r2:0.802
e45, train_loss:0.1331, r2:0.959, val_loss:0.1950, val_r2:0.808
Model successfully saved
e46, train_loss:0.1370, r2:0.962, val_loss:0.1909, val_r2:0.812
```

```
Model successfully saved
e47, train_loss:0.1258, r2:0.963, val_loss:0.1883, val_r2:0.815
Model successfully saved
e48, train_loss:0.1300, r2:0.964, val_loss:0.1866, val_r2:0.816
Model successfully saved
e49, train_loss:0.1253, r2:0.966, val_loss:0.1857, val_r2:0.817
Model successfully saved
e50, train_loss:0.1274, r2:0.968, val_loss:0.1846, val_r2:0.818
Model successfully saved
e51, train_loss:0.1274, r2:0.969, val_loss:0.1822, val_r2:0.821
Model successfully saved
e52, train_loss:0.1250, r2:0.970, val_loss:0.1798, val_r2:0.823
Model successfully saved
e53, train_loss:0.1100, r2:0.971, val_loss:0.1786, val_r2:0.824
Model successfully saved
e54, train_loss:0.1166, r2:0.972, val_loss:0.1784, val_r2:0.824
Model successfully saved
e55, train_loss:0.1199, r2:0.973, val_loss:0.1791, val_r2:0.824
e56, train_loss:0.1179, r2:0.974, val_loss:0.1806, val_r2:0.822
e57, train_loss:0.1187, r2:0.974, val_loss:0.1818, val_r2:0.821
e58, train_loss:0.1076, r2:0.975, val_loss:0.1828, val_r2:0.820
e59, train_loss:0.1054, r2:0.976, val_loss:0.1841, val_r2:0.819
e60, train_loss:0.1192, r2:0.977, val_loss:0.1857, val_r2:0.817
e61, train_loss:0.1258, r2:0.977, val_loss:0.1871, val_r2:0.816
e62, train_loss:0.1106, r2:0.978, val_loss:0.1884, val_r2:0.814
e63, train_loss:0.1114, r2:0.978, val_loss:0.1893, val_r2:0.814
e64, train_loss:0.1055, r2:0.979, val_loss:0.1892, val_r2:0.814
e65, train_loss:0.1163, r2:0.980, val_loss:0.1886, val_r2:0.814
e66, train_loss:0.1090, r2:0.980, val_loss:0.1882, val_r2:0.815
e67, train_loss:0.1193, r2:0.981, val_loss:0.1878, val_r2:0.815
e68, train_loss:0.1186, r2:0.981, val_loss:0.1876, val_r2:0.815
e69, train_loss:0.1089, r2:0.982, val_loss:0.1878, val_r2:0.815
e70, train_loss:0.1160, r2:0.982, val_loss:0.1876, val_r2:0.815
e71, train_loss:0.1031, r2:0.983, val_loss:0.1876, val_r2:0.815
e72, train_loss:0.1093, r2:0.983, val_loss:0.1877, val_r2:0.815
e73, train_loss:0.1061, r2:0.984, val_loss:0.1882, val_r2:0.815
e74, train_loss:0.1080, r2:0.984, val_loss:0.1890, val_r2:0.814
e75, train_loss:0.1053, r2:0.985, val_loss:0.1896, val_r2:0.813
e76, train_loss:0.1002, r2:0.985, val_loss:0.1899, val_r2:0.813
```

```
e77, train_loss:0.1082, r2:0.985, val_loss:0.1901, val_r2:0.813
e78, train_loss:0.1131, r2:0.986, val_loss:0.1901, val_r2:0.813
e79, train_loss:0.1075, r2:0.986, val_loss:0.1903, val_r2:0.812
e80, train_loss:0.0984, r2:0.987, val_loss:0.1908, val_r2:0.812
e81, train_loss:0.1030, r2:0.987, val_loss:0.1912, val_r2:0.812
e82, train_loss:0.1053, r2:0.987, val_loss:0.1915, val_r2:0.811
e83, train_loss:0.1015, r2:0.988, val_loss:0.1917, val_r2:0.811
e84, train_loss:0.1143, r2:0.988, val_loss:0.1921, val_r2:0.811
e85, train_loss:0.1037, r2:0.988, val_loss:0.1926, val_r2:0.810
e86, train_loss:0.1073, r2:0.989, val_loss:0.1931, val_r2:0.810
e87, train_loss:0.1053, r2:0.989, val_loss:0.1934, val_r2:0.809
e88, train_loss:0.1022, r2:0.989, val_loss:0.1941, val_r2:0.809
e89, train_loss:0.0968, r2:0.989, val_loss:0.1949, val_r2:0.808
e90, train_loss:0.1072, r2:0.990, val_loss:0.1958, val_r2:0.807
e91, train_loss:0.1017, r2:0.990, val_loss:0.1969, val_r2:0.806
e92, train_loss:0.0982, r2:0.990, val_loss:0.1977, val_r2:0.805
e93, train_loss:0.0989, r2:0.990, val_loss:0.1982, val_r2:0.805
e94, train_loss:0.0995, r2:0.991, val_loss:0.1985, val_r2:0.804
e95, train_loss:0.1013, r2:0.991, val_loss:0.1987, val_r2:0.804
e96, train_loss:0.0988, r2:0.991, val_loss:0.1987, val_r2:0.804
e97, train_loss:0.1040, r2:0.991, val_loss:0.1987, val_r2:0.804
e98, train_loss:0.1028, r2:0.992, val_loss:0.1988, val_r2:0.804
e99, train_loss:0.0984, r2:0.992, val_loss:0.1991, val_r2:0.804
```

1.4 train_balanced_sampling_sep_mlp_disentangle_cell_type.py（端到端训练 HGAT）

```
e0, train_loss:0.7745, r2:-0.006, val_loss:1.0036, val_r2:0.011
Model successfully saved
e1, train_loss:0.7374, r2:0.029, val_loss:0.9751, val_r2:0.039
Model successfully saved
e2, train_loss:0.7284, r2:0.061, val_loss:0.9476, val_r2:0.067
Model successfully saved
e3, train_loss:0.6925, r2:0.092, val_loss:0.9197, val_r2:0.094
Model successfully saved
e4, train_loss:0.6524, r2:0.124, val_loss:0.8894, val_r2:0.124
Model successfully saved
e5, train_loss:0.6485, r2:0.157, val_loss:0.8558, val_r2:0.157
```

```
Model successfully saved
e6, train_loss:0.6082, r2:0.194, val_loss:0.8195, val_r2:0.193
Model successfully saved
e7, train_loss:0.5922, r2:0.233, val_loss:0.7805, val_r2:0.231
Model successfully saved
e8, train_loss:0.5461, r2:0.274, val_loss:0.7387, val_r2:0.272
Model successfully saved
e9, train_loss:0.5237, r2:0.319, val_loss:0.6937, val_r2:0.317
Model successfully saved
e10, train_loss:0.5139, r2:0.365, val_loss:0.6463, val_r2:0.363
Model successfully saved
e11, train_loss:0.4559, r2:0.413, val_loss:0.5969, val_r2:0.412
Model successfully saved
e12, train_loss:0.4395, r2:0.462, val_loss:0.5468, val_r2:0.461
Model successfully saved
e13, train_loss:0.4196, r2:0.510, val_loss:0.4966, val_r2:0.511
Model successfully saved
e14, train_loss:0.3742, r2:0.557, val_loss:0.4482, val_r2:0.558
Model successfully saved
e15, train_loss:0.3628, r2:0.601, val_loss:0.4035, val_r2:0.603
Model successfully saved
e16, train_loss:0.3431, r2:0.641, val_loss:0.3645, val_r2:0.641
Model successfully saved
e17, train_loss:0.3211, r2:0.678, val_loss:0.3316, val_r2:0.673
Model successfully saved
e18, train_loss:0.2987, r2:0.711, val_loss:0.3050, val_r2:0.700
Model successfully saved
e19, train_loss:0.2923, r2:0.741, val_loss:0.2845, val_r2:0.720
Model successfully saved
e20, train_loss:0.2666, r2:0.767, val_loss:0.2687, val_r2:0.735
Model successfully saved
e21, train_loss:0.2407, r2:0.789, val_loss:0.2558, val_r2:0.748
Model successfully saved
e22, train_loss:0.2505, r2:0.807, val_loss:0.2439, val_r2:0.760
Model successfully saved
e23, train_loss:0.2406, r2:0.823, val_loss:0.2314, val_r2:0.772
Model successfully saved
e24, train_loss:0.2076, r2:0.838, val_loss:0.2158, val_r2:0.787
Model successfully saved
```

```
e25, train_loss:0.2099, r2:0.854, val_loss:0.2019, val_r2:0.801
Model successfully saved
e26, train_loss:0.1825, r2:0.869, val_loss:0.1969, val_r2:0.806
Model successfully saved
e27, train_loss:0.1965, r2:0.881, val_loss:0.2010, val_r2:0.802
e28, train_loss:0.1794, r2:0.890, val_loss:0.2081, val_r2:0.795
e29, train_loss:0.1739, r2:0.899, val_loss:0.2124, val_r2:0.791
e30, train_loss:0.1701, r2:0.908, val_loss:0.2157, val_r2:0.788
e31, train_loss:0.1665, r2:0.915, val_loss:0.2205, val_r2:0.783
e32, train_loss:0.1596, r2:0.917, val_loss:0.2263, val_r2:0.777
e33, train_loss:0.1623, r2:0.921, val_loss:0.2335, val_r2:0.770
e34, train_loss:0.1705, r2:0.926, val_loss:0.2429, val_r2:0.761
e35, train_loss:0.1492, r2:0.931, val_loss:0.2506, val_r2:0.753
e36, train_loss:0.1660, r2:0.934, val_loss:0.2498, val_r2:0.754
e37, train_loss:0.1509, r2:0.937, val_loss:0.2409, val_r2:0.763
e38, train_loss:0.1502, r2:0.942, val_loss:0.2306, val_r2:0.773
e39, train_loss:0.1447, r2:0.945, val_loss:0.2234, val_r2:0.780
e40, train_loss:0.1350, r2:0.947, val_loss:0.2185, val_r2:0.785
e41, train_loss:0.1460, r2:0.949, val_loss:0.2152, val_r2:0.788
e42, train_loss:0.1359, r2:0.953, val_loss:0.2124, val_r2:0.791
e43, train_loss:0.1339, r2:0.955, val_loss:0.2079, val_r2:0.795
e44, train_loss:0.1369, r2:0.957, val_loss:0.2013, val_r2:0.802
e45, train_loss:0.1331, r2:0.959, val_loss:0.1950, val_r2:0.808
Model successfully saved
e46, train_loss:0.1370, r2:0.962, val_loss:0.1909, val_r2:0.812
Model successfully saved
e47, train_loss:0.1258, r2:0.963, val_loss:0.1883, val_r2:0.815
Model successfully saved
e48, train_loss:0.1300, r2:0.964, val_loss:0.1866, val_r2:0.816
Model successfully saved
e49, train_loss:0.1253, r2:0.966, val_loss:0.1857, val_r2:0.817
Model successfully saved
e50, train_loss:0.1274, r2:0.968, val_loss:0.1846, val_r2:0.818
Model successfully saved
e51, train_loss:0.1274, r2:0.969, val_loss:0.1822, val_r2:0.821
Model successfully saved
e52, train_loss:0.1250, r2:0.970, val_loss:0.1798, val_r2:0.823
Model successfully saved
e53, train_loss:0.1100, r2:0.971, val_loss:0.1786, val_r2:0.824
```

```
Model successfully saved
e54, train_loss:0.1166, r2:0.972, val_loss:0.1784, val_r2:0.824
Model successfully saved
e55, train_loss:0.1199, r2:0.973, val_loss:0.1791, val_r2:0.824
e56, train_loss:0.1179, r2:0.974, val_loss:0.1806, val_r2:0.822
e57, train_loss:0.1187, r2:0.974, val_loss:0.1818, val_r2:0.821
e58, train_loss:0.1076, r2:0.975, val_loss:0.1828, val_r2:0.820
e59, train_loss:0.1054, r2:0.976, val_loss:0.1841, val_r2:0.819
e60, train_loss:0.1192, r2:0.977, val_loss:0.1857, val_r2:0.817
e61, train_loss:0.1258, r2:0.977, val_loss:0.1871, val_r2:0.816
e62, train_loss:0.1106, r2:0.978, val_loss:0.1884, val_r2:0.814
e63, train_loss:0.1114, r2:0.978, val_loss:0.1893, val_r2:0.814
e64, train_loss:0.1055, r2:0.979, val_loss:0.1892, val_r2:0.814
e65, train_loss:0.1163, r2:0.980, val_loss:0.1886, val_r2:0.814
e66, train_loss:0.1090, r2:0.980, val_loss:0.1882, val_r2:0.815
e67, train_loss:0.1193, r2:0.981, val_loss:0.1878, val_r2:0.815
e68, train_loss:0.1186, r2:0.981, val_loss:0.1876, val_r2:0.815
e69, train_loss:0.1089, r2:0.982, val_loss:0.1878, val_r2:0.815
e70, train_loss:0.1160, r2:0.982, val_loss:0.1876, val_r2:0.815
e71, train_loss:0.1031, r2:0.983, val_loss:0.1876, val_r2:0.815
e72, train_loss:0.1093, r2:0.983, val_loss:0.1877, val_r2:0.815
e73, train_loss:0.1061, r2:0.984, val_loss:0.1882, val_r2:0.815
e74, train_loss:0.1080, r2:0.984, val_loss:0.1890, val_r2:0.814
e75, train_loss:0.1053, r2:0.985, val_loss:0.1896, val_r2:0.813
e76, train_loss:0.1002, r2:0.985, val_loss:0.1899, val_r2:0.813
e77, train_loss:0.1082, r2:0.985, val_loss:0.1901, val_r2:0.813
e78, train_loss:0.1131, r2:0.986, val_loss:0.1901, val_r2:0.813
e79, train_loss:0.1075, r2:0.986, val_loss:0.1903, val_r2:0.812
e80, train_loss:0.0984, r2:0.987, val_loss:0.1908, val_r2:0.812
e81, train_loss:0.1030, r2:0.987, val_loss:0.1912, val_r2:0.812
e82, train_loss:0.1053, r2:0.987, val_loss:0.1915, val_r2:0.811
e83, train_loss:0.1015, r2:0.988, val_loss:0.1917, val_r2:0.811
e84, train_loss:0.1143, r2:0.988, val_loss:0.1921, val_r2:0.811
e85, train_loss:0.1037, r2:0.988, val_loss:0.1926, val_r2:0.810
```